

CHAPTER: LIMITS AND CONTINUITY

This textbook provides a rigorous approach to calculus fundamentals, covering theoretical proofs, advanced theorems, and long-form analytical problems.

1. Fundamental Theory of Limits

A limit describes the trend of a function $f(x)$ as x approaches a point c . Formally, $\lim_{x \rightarrow c} f(x) = L$ means that for every small $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta$.

One-Sided Limits:

- Left-Hand Limit (LHL): $\lim_{x \rightarrow c^-} f(x)$
- Right-Hand Limit (RHL): $\lim_{x \rightarrow c^+} f(x)$

Existence: A limit exists if and only if $LHL = RHL = L$.

2. Advanced Theorems

The Squeeze Theorem If $g(x) \leq f(x) \leq h(x)$ and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} f(x) = L$.

L'Hopital's Rule Used for indeterminate forms $0/0$ or ∞/∞ :

$$\lim_{x \rightarrow c} f(x)/g(x) = \lim_{x \rightarrow c} f'(x)/g'(x)$$

3. Continuity Framework

A function $f(x)$ is continuous at $x = c$ if:

1. $f(c)$ is defined.
2. $\lim_{(x \rightarrow c)} f(x)$ exists.
3. $\lim_{(x \rightarrow c)} f(x) = f(c)$.

If any condition fails, the function has a discontinuity (removable, jump, or infinite).

4. Long-Answer Problem Bank

Problem 1: Continuity of a Piecewise Function

Analyze the continuity of $f(x)$ at $x=1$, where:

$$f(x) = x^2 + 1 \text{ for } x < 1$$

$$f(x) = 2x \text{ for } x \geq 1$$

Detailed Solution:

1. Find LHL: $\lim_{(x \rightarrow 1^-)} (x^2 + 1) = 1^2 + 1 = 2$.

2. Find RHL: $\lim_{(x \rightarrow 1^+)} (2x) = 2(1) = 2$.

3. Value at point: $f(1) = 2(1) = 2$.

Since $\text{LHL} = \text{RHL} = f(1)$, the function is **continuous** at $x=1$.

Problem 2: Trigonometric Limit derivation using Squeeze Theorem

Prove that $\lim_{(x \rightarrow 0)} (\sin x / x) = 1$.

Detailed Solution:

For small positive x , we have the inequality: $\cos x < (\sin x / x) < 1$.

Applying the Squeeze Theorem as $x \rightarrow 0$:

$$\lim_{(x \rightarrow 0)} \cos x = 1 \text{ and } \lim_{(x \rightarrow 0)} 1 = 1.$$

Therefore, $\lim_{(x \rightarrow 0)} (\sin x / x)$ must be 1.

5. Limits at Infinity

To find $\lim_{x \rightarrow \infty} f(x)/g(x)$, divide every term by the highest power of x in the denominator.

Example: $\lim_{x \rightarrow \infty} (3x^2 + 5) / (2x^2 - 1) = \lim_{x \rightarrow \infty} (3 + 5/x^2) / (2 - 1/x^2) = 3/2$.